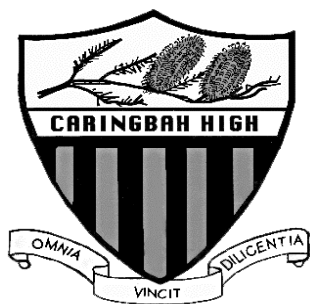




CARINGBAH HIGH SCHOOL

2013

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–5 **10 marks**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–12 **90 marks**

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Which of the following are the foci for the ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

- | | | | |
|-----|--------------------------------|-----|----------------------------|
| (A) | $(0, \pm \frac{3\sqrt{7}}{4})$ | (B) | $(\pm\sqrt{7}, 0)$ |
| (C) | $(0, \pm\sqrt{7})$ | (D) | $(\frac{3\sqrt{7}}{4}, 0)$ |

- 2 Which is the correct answer to the following integral?

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^7(x) \cos^4(x) dx$$

- | | | | |
|-----|--|-----|------------------------------------|
| (A) | $2 \times \int_0^{\frac{\pi}{4}} \sin^7(x) \cos^4(x) dx$ | (B) | $\frac{1024 - 533\sqrt{2}}{36960}$ |
| (C) | $-\frac{1024 - 533\sqrt{2}}{36960}$ | (D) | zero |

- 3 Let α, β and γ be the roots of the cubic equation $x^3 - 5x^2 + 13x - 7 = 0$. Which of the following is the equation with roots α^2, β^2 and γ^2 ?

- | | |
|---------------------------------|---------------------------------|
| (A) $7x^3 - 13x^2 + 5x - 1 = 0$ | (B) $x^3 + x^2 + 99x - 49 = 0$ |
| (C) $x^3 + 5x^2 - 13x - 7 = 0$ | (D) $49x^3 + 99x^2 + x - 1 = 0$ |

4 Given that $x^2 + y^2 + xy = 12$, which of the following is true?

(A) $\frac{dy}{dx} = \frac{2x + y}{2y + x}$ (B) $\frac{dy}{dx} = -\frac{2x + y}{2y + x}$

(C) $\frac{dy}{dx} = \frac{2x - y}{2y + x}$ (D) $\frac{dy}{dx} = \frac{-2x + y}{2y + x}$

5 The equation $|z - 1 - 3i| + |z - 9 - 3i| = 10$ corresponds to an ellipse in the Argand diagram. Which of the following is the complex number corresponding to the centre of the ellipse?

(A) $5 + 3i$ (B) $-5 + 3i$

(C) $-5 - 3i$ (D) $5 - 3i$

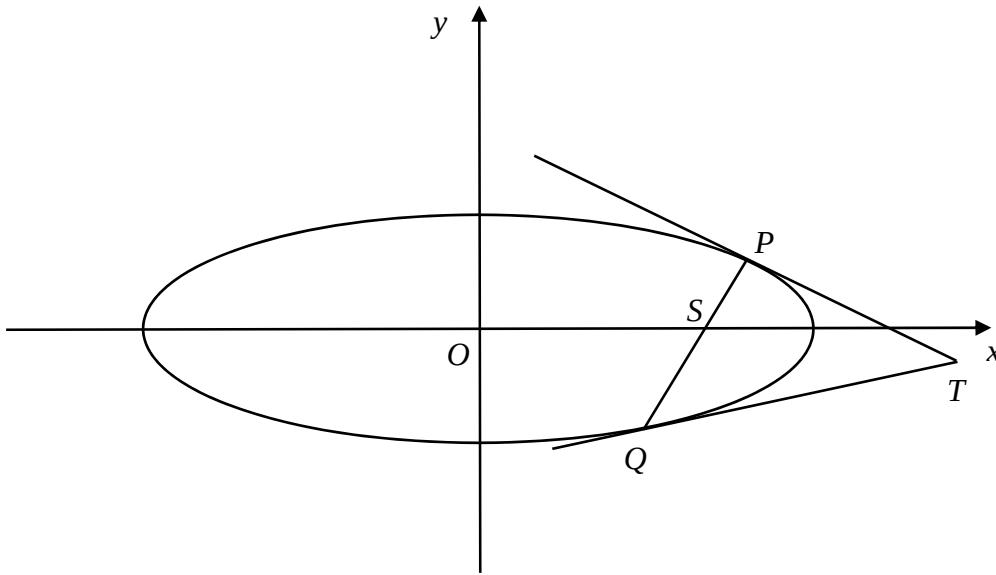
6 The point $T(acos\theta, asin\theta)$ lies on the circle $x^2 + y^2 = r^2$. Which of the following gives the equation of the tangent at T ?

(A) $xcos\theta + ysin\theta = a$ (B) $xcos\theta - ysin\theta = a$

(C) $xcos\theta - ysin\theta = a^2$ (D) $xcos\theta + ysin\theta = a^2$

7

The point P lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The chord through P and the focus $S(ae, 0)$ meets the ellipse at Q . The tangents to the ellipse at P and Q meet at the point $T(x_0, y_0)$, so the equation of PQ is $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$. (Note that T lies on the directrix).



What is the value of the ratio $\frac{PS}{ST}$, given that this ratio is constant?

(A) e^2

(B) ae

(C) $\frac{a}{e}$

(D) e

8 Suppose $\omega^3 = 1$, $\omega \neq 1$ and k is a positive integer.

What are the two values of $1 + \omega^k + \omega^{2k}$?

(A) 3, 0

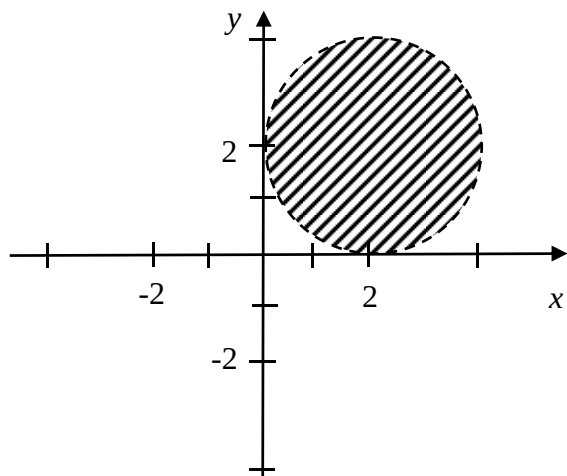
(B) 3, 1

(C) 1, 0

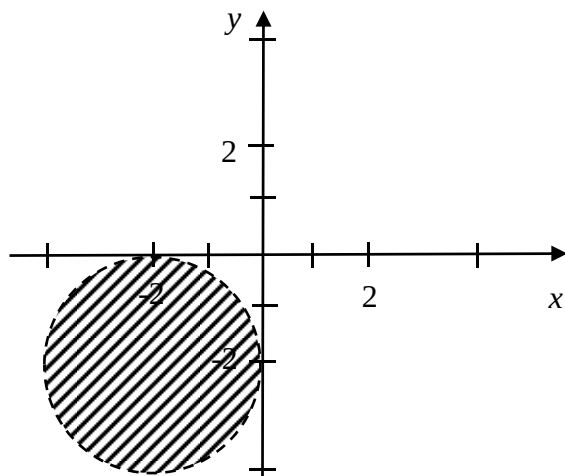
(D) None of the above

9 Which of the following is the graph of $|z - 1 - i| < 2$?

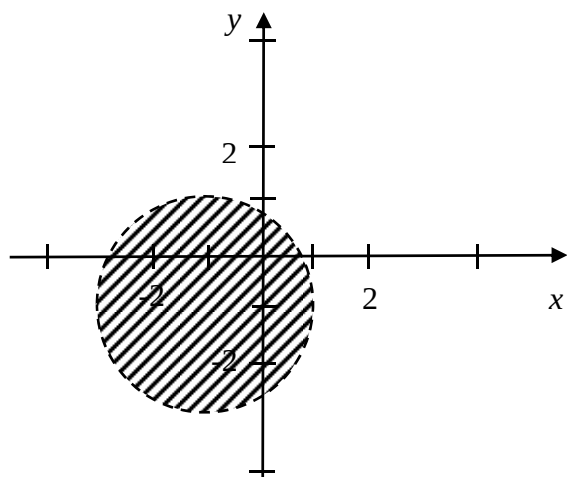
(A)



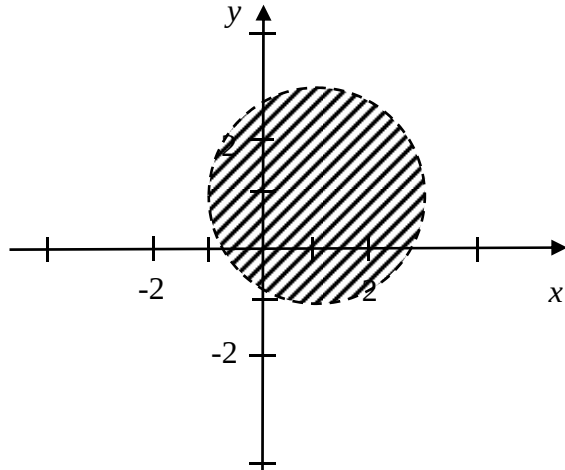
(B)



(C)



(D)



10 Given that $\cos(a + b)x + \cos(a - b)x = 2 \cos(ax) \cos(bx)$, which of the following is the answer for

$$\int \cos(3x) \cos(2x) dx ?$$

(A) $\frac{1}{2}(\cos 5x + \cos x) + c$

(B) $\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + c$

(C) $\frac{1}{10} \sin 5x + \frac{1}{2} \sin x + c$

(D) $\frac{1}{2}(\sin 5x + \sin x) + c$

END OF MULTIPLE CHOICE QUESTIONS

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In

Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.

Marks

(a) Find

(i) $\int \frac{t^2 - 2}{t^3} dt$ 2

(ii) $\int x e^x dx$ 2

(iii) $\int \frac{2x}{(x+1)(x+3)} dx$ 3

(b) By using the substitution $u = x - 4$ evaluate

$$\int_4^{4.5} \frac{dx}{\sqrt{(x-3)(5-x)}}$$
 3

(c) (i) If 3

$$u_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx, \quad n \geq 2$$

prove that

$$u_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1)u_{n-2}$$

(ii) Hence evaluate 2

$$\int_0^{\frac{\pi}{2}} x^2 \sin x \, dx$$

Question 12 (15 marks) Start a NEW booklet.

Marks

- (a) The complex number w is given by $w = -1 + i\sqrt{3}$.
- (i) Show that $w^2 = 2\bar{w}$. 2
- (ii) Evaluate $|w|$ and $\arg w$. 2
- (iii) Show that w is a root of $w^3 - 8 = 0$ 1
- (b) Sketch the locus of z satisfying:
- (i) $\operatorname{Re}(z) = |z|$ 2
- (ii) Both $\operatorname{Re}(z) \geq 2$ and $|z - 1| \leq 2$ 3
- (c) Given that a and b are real numbers and 2
- $$\frac{a}{1+i} + \frac{b}{1+2i} = 1$$
- find the values of a and b .
- (d) The complex numbers z_1, z_2, z_3 and z_4 are represented in the complex plane by the points 3
- A, B, C and D respectively.
- If $z_1 + z_3 = z_2 + z_4$ prove ABCD is a parallelogram.

(a) The equation $x^3 + bx^2 + x + 2 = 0$, where b is a real number, has roots α, β, γ .

(i) Obtain an expression, in terms of b , for 2

$$\alpha^2 + \beta^2 + \gamma^2$$

(ii) Hence determine the set of possible values of b if the roots of the above equation are all real. 1

(iii) Write down the equation whose roots are 2

$$2\alpha, 2\beta, 2\gamma$$

(b) Given that the polynomial $P(x) = 8x^4 - 36x^3 - 66x^2 - 35x - 6$ has a zero of multiplicity 3, find all the zeros of $P(x)$. 3

(c) If z represents a complex number such that $z^5 = 1$, where $z \neq 1$.

(i) Deduce that 2

$$z^2 + z + 1 + \frac{1}{z} + \frac{1}{z^2} = 0$$

(ii) By substituting $x = z + \frac{1}{z}$ reduce the equation in (i) to a quadratic in x . 2

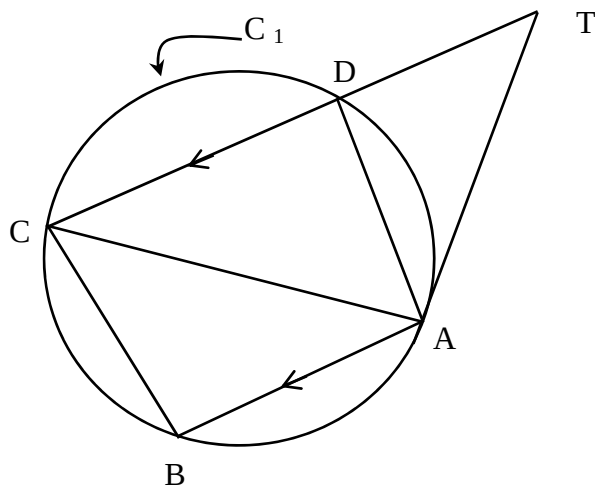
(iii) Hence deduce that 3

$$\cos \frac{2\pi}{5} \cos \frac{4\pi}{5} = -\frac{1}{4}$$

Question 14 (15 marks) Start a NEW booklet.

Marks

- (a) The points A, B, C and D lie on the circle C_1 . From the exterior point T , a tangent is drawn to point A on C_1 . The line CT passes through D and TC is parallel to AB .



- (i) Copy or trace the diagram on to your page.
 (ii) Prove that $\triangle ADT$ is similar to $\triangle ABC$.

3

The line BA is produced through A to point M , which lies on a second circle C_2 . The points A, D, T also lie on C_2 and the line DM crosses AT at Q .

- (iii) Show that $\triangle QMA$ is isosceles.
 (iv) Show that $TM = BC$.

2

2

- (b) (i) Prove that the normal to the hyperbola $xy = 4$ at the point $P(2p, \frac{2}{p})$ is given by

2

$$p^3x - py = 2(p^4 - 1)$$

- (ii) If this normal meets the hyperbola again at $Q(2q, \frac{2}{q})$ prove that $p^3q = -1$.

2

- (iii) Hence prove that there exists only one chord which is normal to the hyperbola at both ends and find its equation.

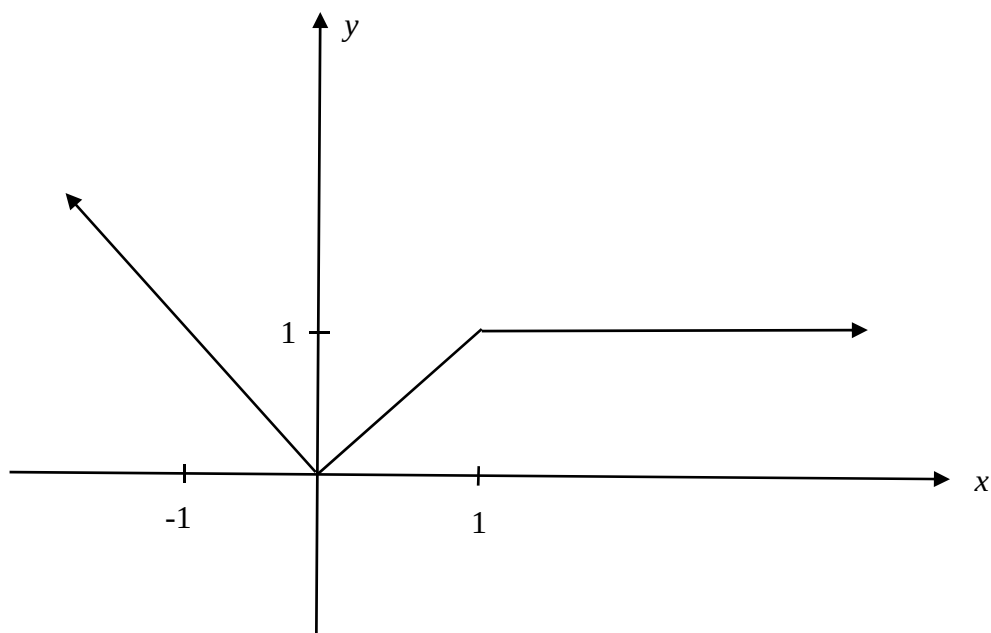
4

Question 15 (15 marks) Start a NEW booklet.

Marks

- (a) Find the equation of the ellipse with centre the origin, which has a focus at $(2,0)$ and the corresponding directrix is $x = 4$. 3

(b)



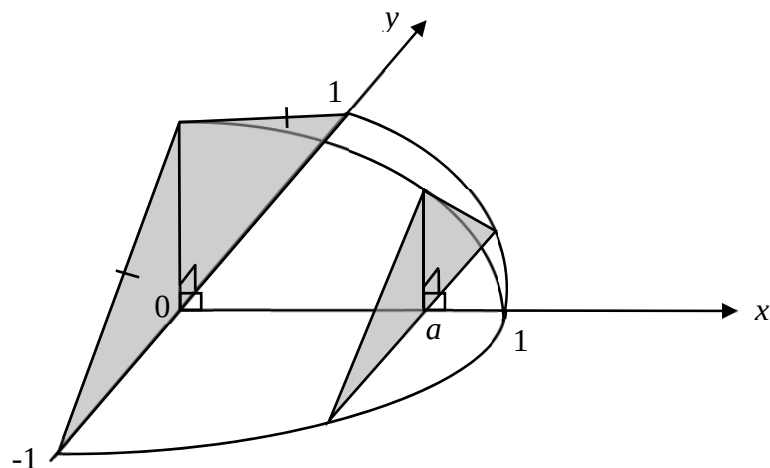
The diagram shows the graph of the function $y = f(x)$

Draw separate sketches of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(-x)$ | 1 |
| (ii) | $y = \frac{1}{f(x)}$ | 1 |
| (iii) | $y = f(x)$ | 1 |
| (iv) | $y = \ln(f(x))$ | 2 |
| (v) | $y = e^{f(x)}$ | 1 |
| (vi) | $x = f(y)$ | 1 |

Question 15 continues on the next page.

- (c) The base of a solid is the semi-circular region of radius 1 unit in the x - y plane as illustrated in the diagram below.



Each cross-section perpendicular to the x -axis is an isosceles triangle with each of the two equal sides three quarters the length of the third side..

- (i) Show that the area of the triangular cross-section at $x = a$ is 2

$$\frac{\sqrt{5}}{2}(1 - a^2).$$

- (ii) Hence find the volume of the solid. 3

- (a) $P(4,6)$ and $Q(14,24)$ are two points on the hyperbola

3

$$\frac{x^2}{4} - \frac{y^2}{12} = 1$$

M is the midpoint of PQ and $O(0,0)$ is the origin. The tangents to the hyperbola at P and Q intersect at the point R . Show that the points R , O and M are collinear.

You may assume that the tangent to this hyperbola at $T(x_1, y_1)$ has equation

$$\frac{x_1 x}{4} - \frac{y_1 y}{12} = 1$$

(Do NOT prove this.)

- (b) A particle is moving so that $\ddot{x} = 18x^3 + 27x^2 + 9x$.

Initially $x = -2$ and the velocity, v , is -6 . It is known that the velocity is always negative.

- (i) Show that $v^2 = 9x^2(1+x)^2$.

2

- (ii) Hence, or otherwise, show that

2

$$\int \frac{1}{x(1+x)} dx = -3t$$

- (iii) Find a, b such that

1

$$\frac{1}{x(1+x)} \equiv \frac{a}{x} + \frac{b}{1+x}$$

- (iv) Show that for some constant c ,

2

$$\log_e \left(1 + \frac{1}{x} \right) = 3t + c$$

- (v) Using this equation and the initial conditions, find x as a function of t .

2

- (c) The angles A, B and C are consecutive terms in an arithmetic series. Show that

3

$$\cos(A) \cos(C) - \cos^2(B) = \sin(A) \sin(C) - \sin^2(B).$$

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

THSC SOLUTIONS

Q1 C	Q6 A
Q2 D	Q7 D
Q3 B	Q8 A
Q4 B	Q9 D
Q5 A	Q10 C

Q11

$$\textcircled{a} \textcircled{1} \int \frac{t^2-2}{t^3} dt = \int \left(\frac{1}{t} - \frac{2}{t^3} \right) dt$$

$$= \int \left(\frac{1}{t} - 2t^{-3} \right) dt$$

$$= \ln|t| + \frac{1}{t^2} + c.$$

$$\textcircled{ii} \int x \cdot e^x dx = \frac{u \cdot v}{x \cdot e^x} - \int \frac{u' \cdot v}{1 \cdot e^x} dx$$

$$= \underline{x e^x - e^x + c}$$

$$\textcircled{iii} I = \int \frac{2x}{(x+1)(x+3)} dx \quad \text{Using partial fractions}$$

$$2x \equiv a(x+3) + b(x+1)$$

$$x = -1 \Rightarrow a = -1$$

$$x = -3 \Rightarrow b = 3$$

$$\therefore I = \int \left(\frac{-1}{x+1} + \frac{3}{x+3} \right) dx$$

$$= 3 \ln|x+3| - \ln|x+1| + c$$

$$= \ln \left| \frac{(x+3)^3}{x+1} \right| + c$$

$$\textcircled{b} I = \int_4^{4.5} \frac{dx}{\sqrt{(x-3)(5-x)}} \quad \text{when } x=4.5, u=0.5$$

$$x=4, u=0$$

$$\text{If } u = x-4$$

$$du = dx$$

$$\int_{u=0}^{u=0.5} \frac{1 \cdot du}{\sqrt{(1+u)(1-u)}}$$

$$u = \frac{1}{2}$$

$$= \int_{u=0}^{\frac{1}{2}} \frac{1}{\sqrt{1-u^2}} du$$

$$= \left[\sin^{-1} u \right]_0^{\frac{1}{2}}$$

$$= \frac{\pi}{6} - 0$$

$$= \underline{\frac{\pi}{6}}$$

$$\textcircled{c} \textcircled{1} U_n = \left[\frac{u \cdot v}{-x^{n-1} \cdot \cos x} \right]_0^{\pi/2} + n \int_0^{\pi/2} x^{n-1} \cdot \cos x \cdot dx$$

$$= 0 + n \int_0^{\pi/2} x^{n-1} \cdot \cos x \cdot dx$$

Now use parts a second time.

$$= n \left[\frac{u \cdot v}{x^{n-1} \cdot \sin x} \right]_0^{\pi/2} - n(n-1) \int_0^{\pi/2} x^{n-2} \cdot \sin x \cdot dx$$

$$= n \left(\left(\frac{\pi}{2} \right)^{n-1} \cdot \sin \frac{\pi}{2} - 0 \right) - n(n-1) U_{n-2}$$

$$= \underline{n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) U_{n-2}}$$

$$\textcircled{ii} n=2 \Rightarrow U_2 = 2 \left(\frac{\pi}{2} \right)^1 - 2 \cdot 1 \cdot U_0$$

$$= \pi - 2 \int_0^{\pi/2} \sin x \cdot dx$$

$$= \pi + 2 \left[\cos x \right]_0^{\pi/2}$$

$$= \pi + 2(0-1)$$

$$= \underline{\pi - 2}$$

Q12 (a)

$$w = -1 + i\sqrt{3}$$

①

$$w^2 = (-1 + i\sqrt{3})(-1 + i\sqrt{3})$$

$$= 1 - 3 - 2i\sqrt{3}$$

$$= -2 - 2i\sqrt{3}$$

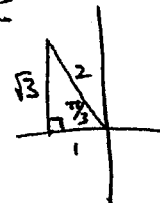
$$\bar{w} = -1 - i\sqrt{3}$$

$$2\bar{w} = 2(-1 - i\sqrt{3})$$

$$= -2 - 2i\sqrt{3}$$

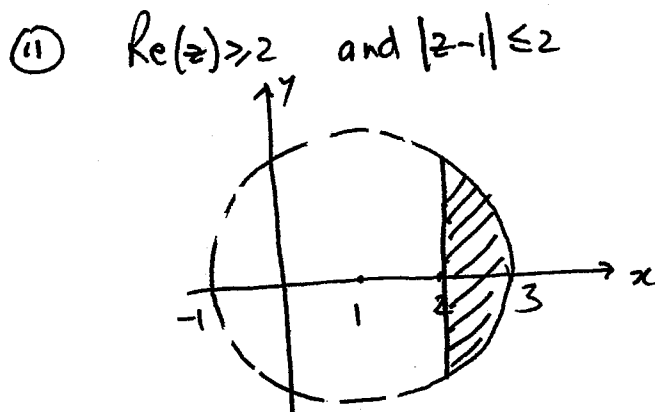
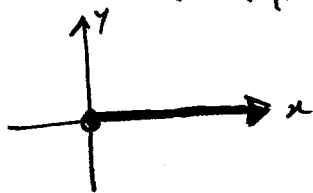
$$= w^2$$

(11) $|w| = \sqrt{1+3} = 2$
 $\arg w = \frac{2\pi}{3}$



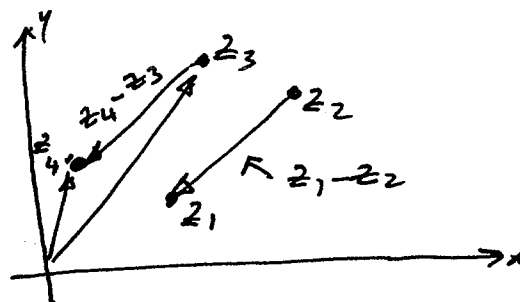
(111) $\therefore w = 2 \operatorname{cis} \frac{2\pi}{3}$
 $w^3 = 2^3 \operatorname{cis} \left(3 \cdot \frac{2\pi}{3}\right)$
 $= 8 \operatorname{cis} 2\pi$
 $= 8 \times 1$
 $w^3 = 8$
 $\therefore w^3 - 8 = 0$

(6) ① Let $z = x + iy$.
 $\operatorname{Re}(z) = x$
 $|z| = \sqrt{x^2 + y^2}$
as $\operatorname{Re}(z) = |z|$
then $x = \sqrt{x^2 + y^2}$
 $x^2 = x^2 + y^2$
 $y^2 = 0$
 $\therefore y = 0$ i.e. x-axis
But $|z| \geq 0 \therefore$ only $x \geq 0$



(c) $\therefore a(1+2i) + b(1+i) = (1+i)(1+2i)$
equating real parts $\Rightarrow a+b = -1$
" imaginary parts $\Rightarrow 2a+b = 3$
 $\therefore \underline{a=4}$
 $\underline{b=-5}$

(d) $\therefore z_1 - z_2 = z_4 - z_3$



If $z_1 - z_2 = z_4 - z_3$
then $|z_1 - z_2| = |z_4 - z_3|$
and $\arg(z_1 - z_2) = \arg(z_4 - z_3)$
 $\therefore AB = CD$
and $AB \parallel CD$
 $\therefore ABCD$ is a parallelogram.

Q13

(a) ① $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$
 $= (-b)^2 - 2 \times 1$
 $= \underline{b^2 - 2}$

② real roots $\Rightarrow \alpha^2 + \beta^2 + \gamma^2 \geq 0$
both $\alpha, \beta, \gamma \neq 0$
 $\therefore \alpha^2 + \beta^2 + \gamma^2 > 0$
 $\therefore b^2 - 2 > 0$
 $\therefore \underline{b < -\sqrt{2} \text{ or } b > \sqrt{2}}$

(iii) $2\alpha = x \therefore \frac{x}{2} = \alpha$
in $\alpha^3 + b\alpha^2 + \alpha + 2 = 0$
 $\therefore \left(\frac{x}{2}\right)^3 + b\left(\frac{x}{2}\right)^2 + \frac{x}{2} + 2 = 0$
 $\frac{x^3}{8} + \frac{bx^2}{4} + \frac{x}{2} + 2 = 0$
 $\therefore \underline{x^3 + 2bx^2 + 4x + 16 = 0}$

(2)

(b) $P'(x) = 32x^3 - 108x^2 - 132x - 35$

$P''(x) = 96x^2 - 216x - 132$

$= 12(8x^2 - 18x - 11)$

$= 12(2x+1)(4x-11)$

$\therefore x = -\frac{1}{2} \text{ or } \frac{11}{4}$

$P(\frac{11}{4}) \neq 0, P(-\frac{1}{2}) = 0$

$\therefore P(x) = (2x+1)^3 Q(x)$
 $= (2x+1)^3 (x-6)$

$\therefore x = -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, 6$

(c) ①

$\therefore z^5 - 1 = 0$

ie $(z-1)(z^4 + z^3 + z^2 + z + 1) = 0$

$z \neq 1$ so $z^4 + z^3 + z^2 + z + 1 = 0$

$z \neq 0$ so $\div$ by z^2

ie $z^2 + z + 1 + \frac{1}{z} + \frac{1}{z^2} = 0$

② $\therefore (z^2 + \frac{1}{z^2}) + (z + \frac{1}{z}) + 1 = 0$

$(z + \frac{1}{z})^2 - 2 + (z + \frac{1}{z}) + 1 = 0$

$\therefore x^2 + x - 1 = 0$

③ From $z^5 = 1$

$z_1 = \text{cis } \frac{2\pi}{5}$

$z_2 = \text{cis } \frac{4\pi}{5}$

$\frac{1}{z_2} = z_3 = \text{cis } \frac{6\pi}{5}$

$\frac{1}{z_1} = z_4 = \text{cis } \frac{8\pi}{5}$

$z_5 = \text{cis } \frac{10\pi}{5} = 1$

These are the
4 roots
of ①

from ④ $x = x_1 \text{ or } x_2$

$x_1 = z_1 + \frac{1}{z_1} = \text{cis } \frac{2\pi}{5} + \text{cis } -\frac{2\pi}{5}$
 $= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$
 $+ \cos -\frac{2\pi}{5} + i \sin -\frac{2\pi}{5}$

as cos EVEN function
and sin ODD function

$\therefore x_1 = 2 \cos(\frac{2\pi}{5})$

and $x_2 = z_2 + \frac{1}{z_2}$
 $= 2 \cos(\frac{4\pi}{5})$

from ① $x_1 \cdot x_2 = -1$

$\therefore 2 \cos(\frac{2\pi}{5}) \cdot 2 \cos(\frac{4\pi}{5}) = -1$

$\therefore \cos(\frac{2\pi}{5}) \cdot \cos(\frac{4\pi}{5}) = -\frac{1}{4}$

Q14

① Let $\angle DAT = \beta$, Let $\angle ABC = \alpha$

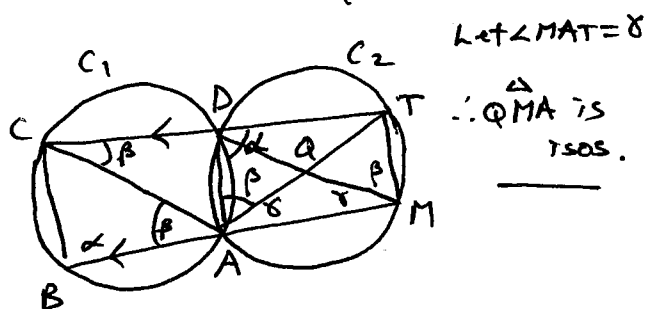
② $\angle DAT = \angle ACD$ ($\angle$ s in alt. seg.)

$\angle ACD = \angle BAC$ (alt $\angle$ s, $CD \parallel AB$)

$\angle ABC = \angle ADT$ (ext $\angle$ cyclic quad $ABCD$)

③ $\angle MAT = \angle MDT$ ($\angle$ s in same seg)

$\angle MDT = \angle AMD$ (alt $\angle$ s, $DT \parallel AM$)



Let $\angle MAT = \delta$

$\therefore \triangle QMA$ is
isos.

④ $\angle DMT = \angle DAT$ ($\angle$ s in same seg)

$\therefore \angle AMT = \beta + \delta$

$\angle ATD = \angle AMD$ ($\angle$ s in same seg)
 $= \gamma$

$\therefore$ From $\triangle ADT$ $\alpha + \beta + \gamma = 180^\circ$

$\therefore \angle BCA = \gamma$

$\therefore \angle BCD = \beta + \delta = \angle AMT$

$\therefore$ BCTM is a parallelogram (1 pair of
sides parallel plus one
pair of opp. $\angle$ s equal).

$\therefore CB = TM$.

(3)

⑥ ①

$$y = \frac{4}{x}$$

$$y' = -\frac{4}{x^2}$$

$$\therefore m_{TP} = -\frac{4}{4p^2}$$

$$= -\frac{1}{p^2}$$

$$m_{NP} = p^2$$

$$\therefore y - \frac{2}{p} = p^2(x - 2p)$$

$$py - 2 = p^3x - 2p^4$$

$$p^3x - py = 2(p^4 - 1) \quad \text{--- (6)}$$

② $\therefore Q$ satisfies $xy = 4$ then $y = \frac{4}{x}$

$$\text{so } p^3 \cdot x - p \cdot \frac{4}{x} = 2(p^4 - 1)$$

$$\text{or } p^3x^2 - 2(p^4 - 1)x - 4p = 0$$

$$\therefore x = \frac{2(p^4 - 1) \pm \sqrt{4(p^4 - 1)^2 + 16p^4}}{2p^3}$$

$$= \frac{p^4 - 1 \pm \sqrt{(p^4 - 1)^2 + 4p^4}}{p^3}$$

$$= \frac{p^4 - 1 \pm (p^4 + 1)}{p^3}$$

$$= \frac{p^4 - 1 + p^4 + 1}{p^3} \text{ or } \frac{p^4 - 1 - p^4 - 1}{p^3}$$

$$= \frac{2p^4}{p^3}, \quad \frac{-2}{p^3}$$

$$= 2p$$

Now if $x = 2p$ then we have $P(2p, \frac{2}{p})$

$$\text{so } Q(2q, \frac{2}{q}) \Rightarrow 2q = \frac{-2}{p^3}$$

$$\text{or } \underline{\underline{p^3q = -1}}$$

③

By symmetry $q^3p = -1$ also

$$\therefore p^3q = p^3q^3$$

$$\text{or } p^3q - q^3p = 0$$

$$pq(p^2 - q^2) = 0$$

$$pq(p - q)(p + q) = 0$$

$$\therefore pq = 0 \text{ --- not possible}$$

$$\text{or } p = q \text{ --- not possible}$$

$$\therefore \underline{p = -q}$$

$$\therefore q^3 - q = -1$$

$$q^4 = 1$$

$$\therefore q = \pm 1$$

$$\therefore p = (\pm 2, \pm 2)$$

$\therefore$ chord is $y = x : -2 \leq x \leq 2$

Q15

(a)

$$ae = 2 \quad \text{--- (1)}$$

$$a/e = 4 \quad \text{--- (2)}$$

$$\text{(1) } \times \text{(2)} \Rightarrow a^2 = 8$$

$$\text{(1)} \div \text{(2)} \Rightarrow e^2 = \frac{1}{2}$$

$$\text{ellipse} \Rightarrow e^2 = 1 - \frac{b^2}{a^2}$$

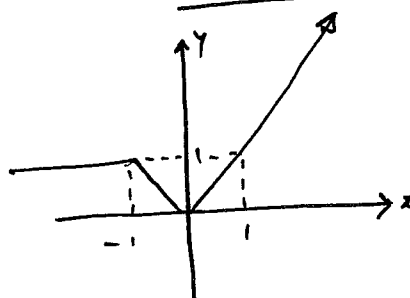
$$\frac{1}{2} = 1 - \frac{b^2}{8}$$

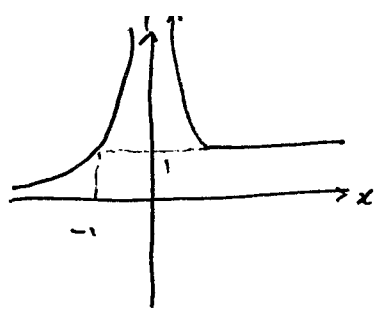
$$4 = 8 - b^2$$

$$\underline{b^2 = 4}$$

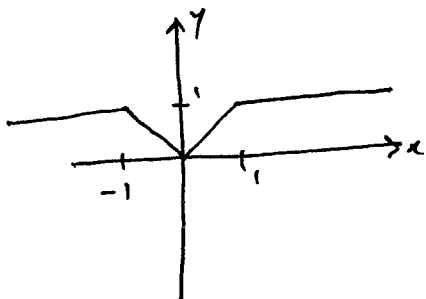
$$\therefore \underline{\underline{\frac{x^2}{8} + \frac{y^2}{4} = 1}}$$

⑥ ①

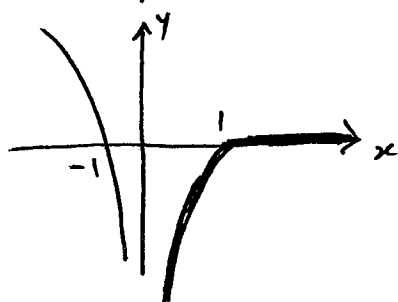




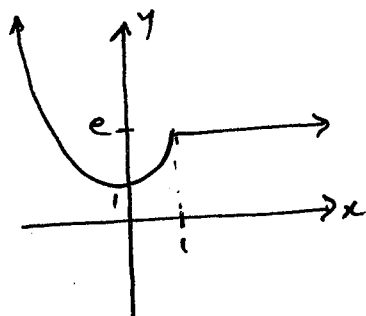
(ii)



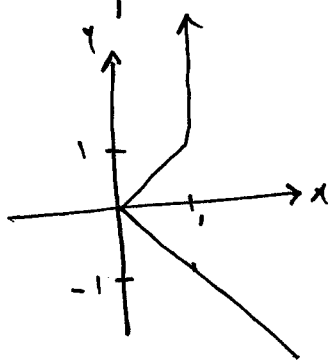
(iv)



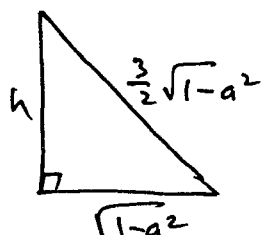
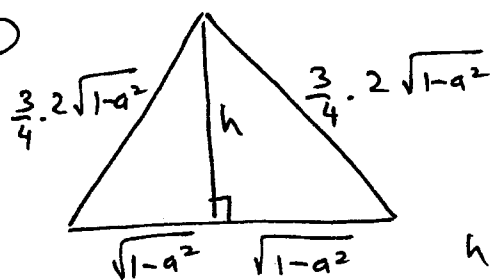
(v)



(vi)



(vii)



$$h^2 = \frac{9}{4}(1-a^2) - (1-a^2)$$

$$= \frac{5}{4}(1-a^2)$$

$$\therefore h = \frac{\sqrt{5}}{2} \sqrt{1-a^2}$$

$$\therefore \text{Area} = \frac{1}{2} (2\sqrt{1-a^2}) \cdot \frac{\sqrt{5}}{2} \sqrt{1-a^2}$$

$$= \frac{\sqrt{5}}{2} (1-a^2)$$

(ii)

$$V = \frac{\sqrt{5}}{2} \int_{x=0}^{x=1} (1-x^2) dx$$

$$= \frac{\sqrt{5}}{2} \left[x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{\sqrt{5}}{2} \left(1 - \frac{1}{3} - (0-0) \right)$$

$$= \frac{\sqrt{5}}{3}$$

Q16 (a) $O \equiv (0,0)$ $M \equiv \left(\frac{18}{2}, \frac{30}{2}\right) \equiv (9,15)$

$$\text{Tangent}_P \equiv T_P \equiv \frac{4x}{4} - \frac{by}{12} = 1$$

$$\therefore x - \frac{y}{2} = 1 \quad \text{--- (1)}$$

$$T_Q \equiv \frac{14x}{4} - \frac{24y}{12} = 1$$

$$\therefore \frac{7x}{2} - 2y = 1 \quad \text{--- (2)}$$

$$\text{(2)} - 4 \times \text{(1)} \quad \frac{7x}{2} - 4x = 1 - 4$$

$$7x - 8x = -6$$

$$-x = -6$$

$$x = 6$$

$$\text{sub into (2)} \quad 6 - \frac{y}{2} = 1$$

$$\frac{y}{2} = 5$$

$$y = 10$$

$$\therefore R \equiv (6,10)$$

$$M_{RO} = \frac{10}{6} = \frac{5}{3}$$

$$M_{MO} = \frac{15}{9} = \frac{5}{3}$$

$$\left. \begin{matrix} M_{RO} = \frac{5}{3} \\ M_{MO} = \frac{5}{3} \end{matrix} \right\} M_{RO} = M_{MO} \Rightarrow RMO \text{ collinear}$$

(5)

$$(i) \int \frac{d}{dx} \frac{1}{2} v^2 dx = \int (18x^3 + 27x^2 + 9x) dx$$

$$\frac{1}{2} v^2 = \frac{9x^4}{2} + 9x^3 + \frac{9x^2}{2} + C$$

$$v^2 = 9x^4 + 18x^3 + 9x^2 + C'$$

$$x = -2, v = -6 \therefore C' = 0$$

$$\therefore v^2 = 9x^2(x^2 + 2x + 1) \\ = 9x^2(x+1)^2$$

$$(ii) \therefore v = 3x(1+x) \text{ or } -3x(1+x)$$

$$\text{here } x = -2 \Rightarrow v = -6x - 1 \quad \left| \begin{array}{l} \text{here} \\ x = -2 \\ v = 6x - 1 \\ = -6 \end{array} \right.$$

$$\therefore v = -3x(1+x)$$

$$\therefore \frac{dx}{dt} = -3x(1+x)$$

$$\frac{dt}{dx} = -\frac{1}{3} \frac{1}{x(1+x)}$$

integrate both sides w.r.t x gives

$$t = -\frac{1}{3} \int \frac{1}{x(1+x)} dx$$

$$\text{or } -3t = \int \frac{1}{x(1+x)} dx$$

$$(iii) 1 \equiv a(1+x) + bx$$

$$x=0 \Rightarrow a=1$$

$$x=-1 \Rightarrow b=-1$$

$$(iv) \therefore -3t = \int \left(\frac{1}{x} - \frac{1}{1+x} \right) dx \\ = \ln|x| - \ln|1+x| + C \\ 3t = \ln \left| \frac{1+x}{x} \right| + C \\ 3t + C' = \ln \left| 1 + \frac{1}{x} \right|$$

(v)

$$t=0 \quad x=-2$$

$$\therefore \ln\left(\frac{1}{2}\right) = C'$$

$$\therefore 1 + \frac{1}{x} = e^{3t + \ln \frac{1}{2}} \\ = \frac{1}{2} e^{3t}$$

$$\frac{1}{x} = \frac{1}{2} e^{3t} - 1$$

$$x = \frac{1}{\frac{1}{2} e^{3t} - 1}$$

$$x = \frac{2}{e^{3t} - 2}$$

(c) AP $\Rightarrow A, B, C \equiv B-d, B, B+d$
so we need only show.

$$\cos A \cos C - \sin A \sin C = \cos^2 B - \sin^2 B \\ = \cos(2B)$$

$$\text{LHS} = \cos(A+C)$$

$$= \cos(B-d + B+d)$$

$$= \cos(2B)$$

$$= \underline{\underline{\text{RHS}}}$$